1 Gain law

(bars denote normalized quantities: lengths by R, $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $0 < \bar{a} < 1$, $1{+}\delta a > 0$; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}'(\bar{a},\, \delta a) := \frac{\mathrm{d}\bar{a}}{\mathrm{d}\bar{w}} = \big(1-\bar{a}\big)^2\big(1+\delta a\big) \qquad \left(\begin{array}{l}\text{no parameter in the law;}\\ \delta a \text{ is a state}\end{array}\right)$$

$$\begin{aligned}\bar{a}'_w[k] &= \big(1-\bar{a}_w[k]\big)^2\big(1+\delta a_w[k]\big)\\ \hat{\bar{a}}'[k] &= \big(1-\hat{\bar{a}}_w[k]\big)^2\big(1+\delta\hat{a}_w[k]\big) \qquad \text{(states only: no } \bar{w}_d,\ \text{no } \hat{\bar{w}}_s\text{)}\end{aligned}$$

Exponent $2 = 1 + 1/p$ at $p = 1$; both asymptotic anchors give $p = 1$, so the exponent is anchored, not chosen.

Integrated form (used for seeding only; $\bar{w}_0$ = integration constant):

$$\bar{a}(\bar{w}) = 1 - \frac{1}{(1+\delta a)\,(\bar{w}-\bar{w}_0)}, \qquad \frac{\partial \bar{a}}{\partial \bar{w}_0} = -\bar{a}' \quad \text{(exact)}$$

$$\text{plane, } \bar{w}_0 = \bar{w}_s - 1: \quad \bar{a} = 1 - \frac{b}{\bar{w}}, \qquad b := \frac{1}{1+\delta a}, \qquad b \to \tfrac{9}{8} \iff \delta a \to -\tfrac{1}{9}$$

$$a_w = a_o\,\bar{a}_w, \qquad a_o = \frac{\Delta t}{\gamma_N R} \qquad (a_o \text{ fixed, not a state})$$

Declared truncation (shape floor; $b_{\text{state}}(\bar{w}) := (1-\bar{a}_{\text{true}})^2/\bar{a}'_{\text{true}}$, offline):

$$\sup_{\bar{w}}\left|\frac{\bar{a}'(\bar{a}_{\text{true}},\delta a)}{\bar{a}'_{\text{true}}} - 1\right| = \sup_{\bar{w}}\left|\frac{b_{\text{state}}(\bar{w})}{b} - 1\right| \qquad \begin{array}{l}0.66\,\%\ \text{RMS},\ 4.65\,\%\ \sup \quad (\bar{w}-\bar{w}_s \in [0.1,\, 25])\\ 0.59\,\%\ \text{RMS},\ 2.06\,\%\ \sup \quad (\bar{w}-\bar{w}_s \geq 1)\end{array}$$

2 Gain rate and parameter Jacobians

$$\begin{aligned}A_a(\bar{a},\delta a) &:= \frac{\partial \bar{a}'}{\partial \bar{a}} = -2\big(1-\bar{a}\big)\big(1+\delta a\big) = \frac{-2\,\bar{a}'}{1-\bar{a}} \qquad (< 0 \text{ always})\\ A_\delta(\bar{a}) &:= \frac{\partial \bar{a}'}{\partial \delta a} = \big(1-\bar{a}\big)^2 = \frac{\bar{a}'}{1+\delta a} \qquad (> 0 \text{ always; no zero, no sign change})\end{aligned}$$

$$\bar{a}'_w[k] - \hat{\bar{a}}'[k] = A_a\big(\hat{\bar{a}}_w[k], \delta\hat{a}_w[k]\big)\, e_{\bar{a}_w}[k] + A_\delta\big(\hat{\bar{a}}_w[k]\big)\, e_{\delta a_w}[k] + \mathcal{O}(e^2)$$

$\bar{a}'$ depends on the state $\bar{a}_w$: $A_a \neq 0$. This is the single structural difference from the height writing and it propagates into S6 row 4 and S7.

3 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k-d+j-1],\ j=1,\dots,d+1,\ d=2;\ \delta\bar{w}_D \equiv 0 \text{ hereafter}\big)$

$$\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\bar{a}_w[k+1] &= \bar{a}_w\big(\bar{w}[k+1]\big) \\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta\bar{w}_d[k] \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \qquad (e_{\bar{a}_w} = \bar{a}_w - \hat{\bar{a}}_w) \\
\delta a_w[k+1] &= \delta a_w[k]
\end{aligned}$$

$$\begin{aligned}
F_{dw}[k] &:= \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{f}_{dw}[k-i] \\
\varepsilon_w[k] &:= \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{a}_w[k-i]\,\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k]
\end{aligned}$$

4 Process model

$$\bar{w}[k+1] - \bar{w}[k] = \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]$$

$$\bar{a}_w[k+1] = \bar{a}_w[k] + \bar{a}'[k]\Big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]\Big]$$

$$\bar{a}'[k] = \big(1-\bar{a}_w[k]\big)^2\big(1+\delta a_w[k]\big)$$

$$\mathbf{x}[k] = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ \delta a_w\,\big]^\top[k] \qquad (d=2)$$

$$\begin{aligned}
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'[k]\Big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]\Big] \\
\delta a_w[k+1] &= \delta a_w[k]
\end{aligned}$$

5 Estimator

$$
\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \hat{\bar{a}}'[k]\,M[k] \\
\delta\hat{a}_w[k+1] &= \delta\hat{a}_w[k]
\end{aligned}
$$

$$
\begin{aligned}
M[k] &:= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}'[k] &= \big(1-\hat{\bar{a}}_w[k]\big)^2\big(1+\delta\hat{a}_w[k]\big)
\end{aligned}
$$

6 Error dynamics

(True − Estimator; coefficients at the estimate)

$$
e_{\delta\bar{w}_j} = \delta\bar{w}_j - \delta\hat{\bar{w}}_j \quad (j=1,2,3), \qquad \mathbf{e} = \big[\, e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_{\bar{a}_w},\ e_{\delta a_w}\,\big]^\top
$$

$$
\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\,e_{\delta\bar{w}_3}[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \Big[\,1 + \hat{\bar{a}}'[k]\,F_{dw}[k] + A_a[k]\,M[k]\,\Big] e_{\bar{a}_w}[k] + A_\delta[k]\,M[k]\ e_{\delta a_w}[k] \\
&\qquad + (1-\lambda_c)\,\hat{\bar{a}}'[k]\,e_{\delta\bar{w}_3}[k] + \hat{\bar{a}}'[k]\,\varepsilon_w[k] \\
e_{\delta a_w}[k+1] &= e_{\delta a_w}[k]
\end{aligned}
$$

$$
A_a[k] = A_a\big(\hat{\bar{a}}_w[k], \delta\hat{a}_w[k]\big), \qquad A_\delta[k] = A_\delta\big(\hat{\bar{a}}_w[k]\big), \qquad \mathbf{e}[k+1] = F_e[k]\,\mathbf{e}[k] + \mathbf{q}[k]
$$

$$
F_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & \lambda_c & -F_{dw} & 0 \\
0 & 0 & (1-\lambda_c)\hat{\bar{a}}' & 1+\hat{\bar{a}}'F_{dw}+A_aM & A_\delta M \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}
$$

$$
\mathbf{q}[k] = \big[\,0,\ 0,\ -\varepsilon_w[k],\ \hat{\bar{a}}'[k]\,\varepsilon_w[k],\ 0\,\big]^\top \qquad (q_4 = -\hat{\bar{a}}'[k]\,q_3)
$$

The term A_aM has no counterpart in the height writing (there $F_e(4,4) = 1 + \hat{\bar{a}}'F_{dw}$ exactly). Homogeneous propagation of the integration constant is exact:

$$
e_{\bar{a}}(\bar{w}) = -\,\bar{a}'(\bar{w})\ e_{\bar{w}_0} \ \propto\ \frac{1}{(\bar{w}-\bar{w}_0)^2} \qquad \Longrightarrow \qquad \oint \text{(closed excursion)} : \text{gain} = 1
$$

7 Measurement and innovation

Measurement (d-step delay; $\bar{a}_{wm} = a_{wm}/a_o$; y_2 raw readout — AR(1) whitening enters with R_2, S8)

$$
\begin{aligned}
\nabla_d \bar{w}_d[k] &:= \bar{w}_d[k] - \bar{w}_d[k-d] \\
y_1[k] &= \delta\bar{w}_m[k] = \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_{wm}[k] = \bar{a}_w[k-d] + n_a[k]
\end{aligned}
$$

Delay back-off

$$\bar{a}_w[k-d] = \bar{a}_w[k] - \bar{a}'[k]\,\nabla_d\bar{w}_d[k] \implies y_2[k] = \bar{a}_w[k] - \bar{a}'[k]\,\nabla_d\bar{w}_d[k] + n_a[k]$$

Linearization (partials at the estimate; $\bar{a}'$ *does* depend on the state $\bar{a}_w$, so $\partial y_2/\partial\bar{a}_w$ is *not* 1 — this is where the height-writing derivation does not carry over)

$$
\begin{aligned}
\frac{\partial y_2}{\partial \bar{a}_w} &= 1 - A_a[k]\,\nabla_d\bar{w}_d[k] = 1 + 2\big(1 - \hat{\bar{a}}_w\big)\big(1 + \delta\hat{a}_w\big)\nabla_d\bar{w}_d[k] \\
\frac{\partial y_2}{\partial \delta a_w} &= -A_\delta[k]\,\nabla_d\bar{w}_d[k] = -\big(1 - \hat{\bar{a}}_w\big)^2\,\nabla_d\bar{w}_d[k]
\end{aligned}
$$

Output equation (H is the Jacobian; row 2 is nonlinear in $\mathbf{x}$, so $H\mathbf{x}$ is its linearization, not an identity)

$$\mathbf{y}[k] = H[k]\,\mathbf{x}[k] + \mathbf{r}[k], \qquad \mathbf{r}[k] = \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 - A_a\nabla_d\bar{w}_d & -A_\delta\nabla_d\bar{w}_d \end{bmatrix}$$

Innovation

$$
\begin{aligned}
e_{y_1}[k] &= y_1[k] - \delta\hat{\bar{w}}_1[k] = e_{\delta\bar{w}_1}[k] + n_w[k] \\
e_{y_2}[k] &= y_2[k] - \hat{\bar{a}}_w[k] + \hat{\bar{a}}'[k]\,\nabla_d\bar{w}_d[k] \\
&= \big[\,1 - A_a[k]\,\nabla_d\bar{w}_d[k]\,\big]\,e_{\bar{a}_w}[k] - A_\delta[k]\,\nabla_d\bar{w}_d[k]\;e_{\delta a_w}[k] + n_a[k]
\end{aligned}
$$

8 Process and measurement noise

Process noise Q (only the current white thermal step enters Q: the past thermal steps in ε_w are cross-step correlated, and n_w already sits in R_1)

$$q_3[k] = -\bar{a}_w[k]\,\bar{f}_T[k], \qquad q_4[k] = \hat{\bar{a}}'[k]\,\bar{a}_w[k]\,\bar{f}_T[k] = -\hat{\bar{a}}'[k]\,q_3[k]$$

$$\gamma(\bar{w}) = \gamma_N c(\bar{w}), \quad \sigma^2_{\bar{f}_T} = a_o^2 \frac{4k_BT\gamma(\bar{w})}{\Delta t}, \quad \bar{a}_w = \frac{\Delta t}{a_o\gamma(\bar{w})R} \implies \mathrm{Var}\big(\bar{a}_w\bar{f}_T\big) = \frac{4k_BT}{a_oR}\,\bar{a}_w$$

$$Q_{33}[k] = \frac{4k_BT}{a_oR}\,\bar{a}_w[k], \quad Q_{34}[k] = Q_{43}[k] = -\hat{\bar{a}}'[k]\,Q_{33}[k], \quad Q_{44}[k] = \hat{\bar{a}}'[k]^2\,Q_{33}[k]$$

$$Q_{55} = 0 \qquad (\delta a_w \text{ constant: no stochastic driving})$$

All other entries vanish; the gain block is rank one, since $q_4 = -\hat{\bar{a}}'[k]\,q_3$. (run time: entries at the current estimate)

Measurement noise $R = \mathrm{diag}(R_1, R_2)$ (a_{cov} = EWMA coefficient, IF_{var} = MA(d) inflation factor of $\hat{\sigma}^2_{\delta\bar{w}_r}$; y_2 treated as white here)

$$R_1 = \sigma^2_{n_w}$$

$$\sigma^2_{\delta\bar{w}_r}[k] = C_{dpmr}\frac{4k_BT}{a_oR}\bar{a}_w[k] + C_n\sigma^2_{n_w} = C_{dpmr}\frac{4k_BT}{a_oR}\big(\bar{a}_w[k] + \xi\big), \qquad \xi := \frac{C_n\sigma^2_{n_w}}{C_{dpmr}\big(4k_BTa_o/R\big)}$$

$$\mathrm{Var}\big(\hat{\sigma}^2_{\delta\bar{w}_r}\big) = K_{\mathrm{var}}IF_{\mathrm{var}}\big(\sigma^2_{\delta\bar{w}_r}\big)^2, \quad K_{\mathrm{var}} = \frac{2a_{\mathrm{cov}}}{2 - a_{\mathrm{cov}}}, \quad R_2[k] = K_{\mathrm{var}}IF_{\mathrm{var}}\big(\bar{a}_w[k] + \xi\big)^2 + d\,Q_{44}[k]$$

Seed, prior and the covariance identity

Seed ($\hat{\bar{w}}_0$ = nominal wall; it enters here and nowhere else)

$$\delta\hat{a}_w[0] = 0, \qquad \hat{\bar{a}}_w[0] = 1 - \frac{1}{\big(1 + \delta\hat{a}_w[0]\big)\big(\bar{w}[0] - \hat{\bar{w}}_0\big)}$$

Level Jacobians at the seed (exact sensitivities of the flow, both at fixed $\bar{w}$)

$$\left.\frac{\partial\bar{a}}{\partial\,\delta a}\right|_{\bar{w}} = \frac{1 - \bar{a}}{1 + \delta a}, \qquad \left.\frac{\partial\bar{a}}{\partial\bar{w}_0}\right|_{\bar{w}} = -\,\bar{a}'$$

$$P_{44}[0] = \left(\frac{1 - \hat{\bar{a}}_w}{1 + \delta\hat{a}_w}\right)^2 P_{\delta a} \;+\; \big(\hat{\bar{a}}'\big)^2 P_{\bar{w}_0} \;+\; \mathrm{floor}_a^2$$

$$P_{45}[0] = \frac{1 - \hat{\bar{a}}_w}{1 + \delta\hat{a}_w}\,P_{\delta a}, \qquad P_{55}[0] = P_{\delta a}$$

$P_{\bar{w}_0}$ is the wall-position prior carried over unchanged; there is no $\bar{w}_s$ state to hold it. $P_{\delta a}$ is uninformative (user decision (i), 2026-08-11): the only undetermined number in the derivation, and the one that makes

$\delta\hat{a}_w[\infty] \to -\frac{1}{9}$ an *acceptance*, not an input.

Covariance identity (predict-only; both terms are exact, so P_{44} must reproduce them with no free constant)

$$P_{44}[k] = \left(\bar{a}'[k]\right)^2 P_{\bar{w}_0} + \left(\frac{1-\bar{a}[k]}{1+\delta a}\right)^2 P_{\delta a} + \mathrm{floor}_a^2$$